

Date: May. 13th. 2021

Specification for Approval

Customer Name : LENOVO**Product Name : 12.6" WQHD AMOLED****Model Name : AMSA26ZP01-0****Description : 12.6" WQHD(2560*1600) 16.7M Color**

Proposed by			Customer's Approval
Designed	Checked	Approved	
HD CHOI	YJ NAM	WJ LEE	

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The specification on this document is going to apply to Mass production according to an agreement which Lenovo and SDC make through the signature on the document

This signature on this document indicates that Lenovo and SDC confirm all items specified in this document have been verified under Display Module as well as Tablet set condition at both sides.

Model No.: AMSA26ZP01	SAMSUNG Display CO., LTD. (All rights reserved.)			
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• REVISION HISTORY

Date.	Rev.No.	Page	Revision Description
21/05/13	000	All	Initial version
21/05/31	001	- 27 - 30 - 30 - 31 - 32 - 33	- ELA, Color Difference, Random Mura Gray level modification : @ 87G → @127G - Cover Panel cutoff area item added - Window (IR LED,Smart Vision,TOF, ALS Sensor) added - Part of C-PCB items was deleted "When COVER PCB TAPE is out of Panel edge, it is NG" C-PCB lifting at COF Bending area was added - Hook item was added - FPC (no#19) was deleted (Same as #10 in page30)

Preliminary

1. GENERAL DESCRIPTION

DESCRIPTION

The AMSA26ZP01-0 uses a organic light emitting diode display (OLED) that uses low temperature poly silicon TFTs as switching components. This model is composed of a TFT OLED panel, a driver circuit.

This 12.6" model has a resolution of 2560 x 1600 pixels and can display up to 16.7M colors.

FEATURES

OLED Display

High contrast ratio & Fast Response

WQHD(2560 x 1600 pixels) resolution

MIPI-DSI 4lanes, 1.6Gbps/lane(MAX)

APPLICATIONS

Tablet PC

If the intent to use this product is for other purpose, please contact Samsung Display.

GENERAL INFORMATION

Item	Specification	Unit	Note
Display area	271.411 (H) x 169.632 (V) (12.6" diagonal)	mm	
Driver Element	LTPS TFT active matrix		
Display colors	16.7M		
Number of pixel	2560 * 1600 (WQHD)	Pixel	16:10
Pixel Arrangement	RGB Stripe Type		
Pixel pitch	106.02 x 106.02, Sub Pixel : 35.34 x 106.02	um	
Display Mode	OLED		
Touch	On Cell Touch		
Thickness of glass	0.6 (Encap 0.4 + LTPS 0.2)	mm	
Surface treatment	Glare		

MECHANICAL INFORMATION

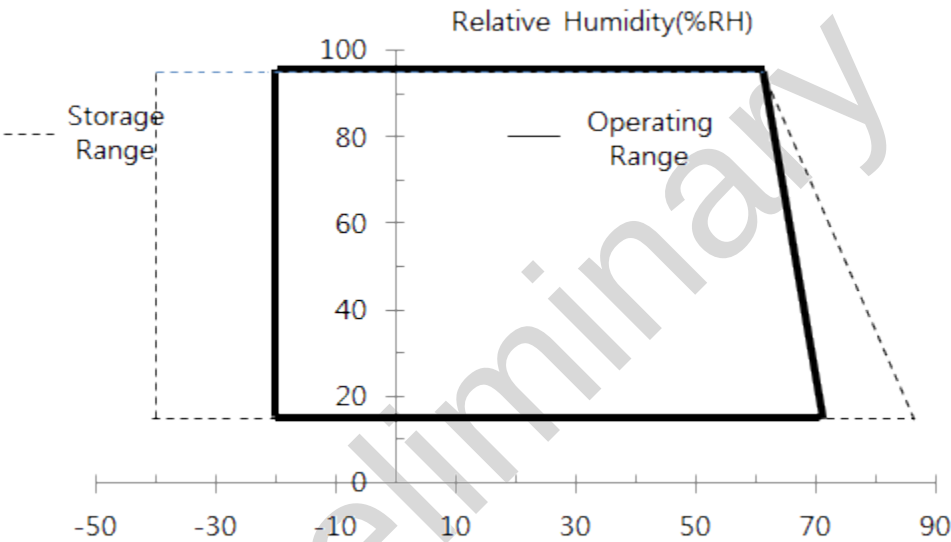
Item		Min.	Typ.	Max.	Unit	Note
Module Size	Horizontal (H)	283.31	283.41	283.51	mm	-
	Vertical (V)	182.23	182.33	182.43	mm	-
	Depth (D)	-	(1.607)	-	mm	Window to Cushion
Weight		-	(160)	-	g	w/o Protective, etc. Film

2. ABSOLUTE MAXIMUM RATINGS

2.1 ENVIRONMENTAL ABSOLTE RATINGS

Item	Symbol	Min.	Max.	Unit	Note
Storage temperature	TSTG	-40	85	°C	(1)
Operating temperature (Tem- perature of Environment)	TOPR	-20	70	°C	(1)

Note (1) The range of temperature and relative humidity are shown in the graph below 93% RH.



2.2 ELECTRICAL ABSOLUTE RATINGS

(1) OLED MODULE

$$V_{SS} = GND = 0V$$

Item		Symbol	Min.	Max.	Unit	Note
Supply Voltage	System	VDD_3.3V	-0.3	+4.6	V	(1), (2)
	EL Power	ELVDD	-0.3	+6.0	V	(1), (2)
		ELVSS	-9.0	+0.3	V	(1), (2)
	TSP	TSP_AVDD	-0.3	+3.9	V	(1), (2)
		TSP_DVDD	-0.3	+3.9	V	(1), (2)
Signal Input Voltage		System_I/O	-0.3	VDD_1.8V+0.2	V	(2)

Note (1) VDD_3.3V, VBAT should satisfy the condition of VDD_3.3V, VBAT > VSS(AGND).

Note (2) If the supplied voltage exceeds the maximum limitation, D-IC can be damaged permanently. Therefore, while operating, it's recommended to use D-IC within the maximum electrical limitation. If not, D-IC can cause decreased reliability or operational problems.

Note (3) Temperature should not exceed 29°C and there should be no condensation.

2.3 THE OTHERS

(1) STATIC ELECTRICITY PRESSURE RESISTANCE

Item	Test Conditions	Remark
CONTACT DISCHARGE	Edge ±6kv Center ±8kv, 150pF, 330Ω, Front 5point	Non-Operating
AIR DISCHARGE	Edge ±8kv Center ±15kv, 150pF, 330Ω, Front 5point	Non-Operating

3. OPTICAL CHARACTERISTICS

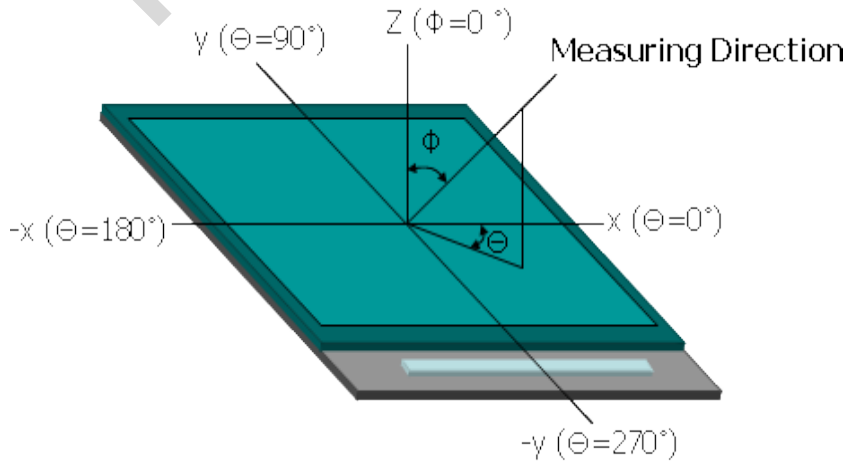
The following items are measured under the stable conditions.* The optical characteristics should be measured in the dark room or the equivalent environment by the methods shown in the Note (5).

Measuring equipment : TOPCON SR-UL2 @120Hz,

$T_a = 25 \pm 2^\circ\text{C}$, $V_{\text{LCD_VCC}} = 3.3\text{V}$, $f_v = 120\text{Hz}$

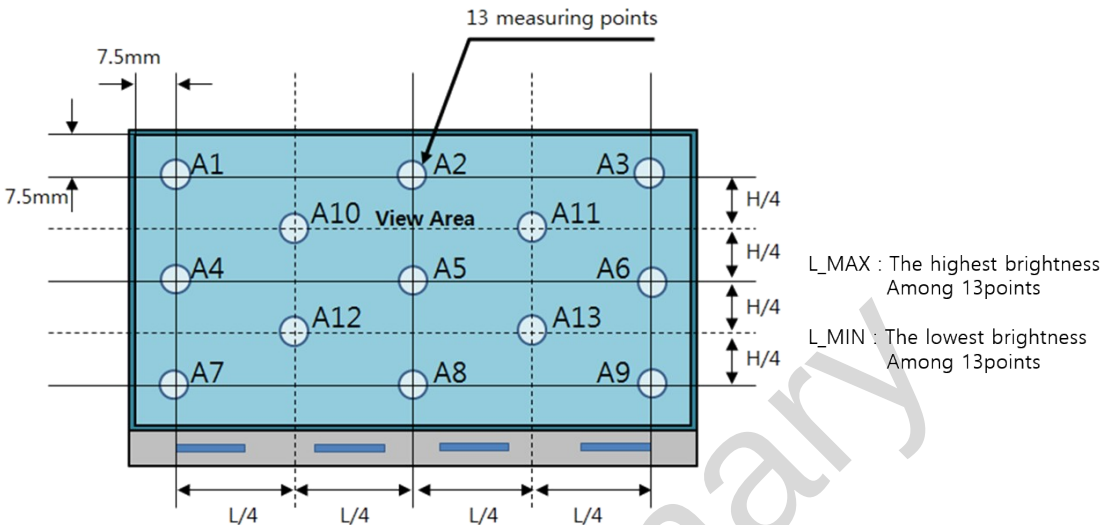
Item		Symbol	Temp	Condition	Min.	Typ.	Max.	Unit	Note
Brightness			25°C	Normal (White Mode)	378	420	462	cd/m²	
Uniformity			25°C	Normal (White Mode)	75	85	-	%	(2)
Contrast ratio		K	25°C	Φ=0°,θ=0°	60,000	-	-	-	(3)
Color of CIE coordinate	White	x	25°C	Φ=0° θ=0°	0.275	0.295	0.315	-	(1), (4)
		y			0.295	0.315	0.335	-	
	Red	x			0.668	0.680	0.692	-	
		y			0.307	0.319	0.331	-	
	Green	x			0.197	0.237	0.277	-	
		y			0.695	0.723	0.751	-	
	Blue	x			0.125	0.138	0.151	-	
		y			0.030	0.050	0.070	-	
Color Gamut			25°C	DCI P3	90	100	-	%	(1), (4)
White Temperature °K			25°C	Φ=0° θ=0°	-	7800	-		(1), (4)
GAMMA			25°C	Φ=0° θ=0° @Full white	2.0	2.2	2.4	-	(1), (4)
Response Time			25°C	On/Off	-	1	-	ms	(1), (4)
View angle			25°C	Upper/Down/Right/Left Contrast Ratio ≥10	Over 80°				(4)
WAD		Δuv	25°C	Φ=45°(@ 9 o'clock)	0.022				(1)

NOTE (1) Definition of Φ , θ



NOTE (2) Uniformity measuring point

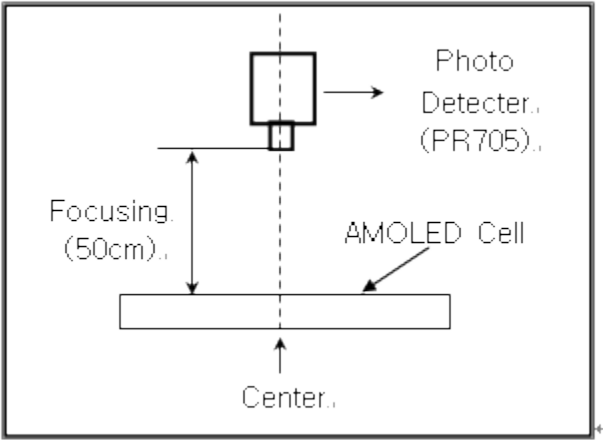
$$Uniformity = \frac{L_{min}}{L_{max}} \times 100 [\%]$$



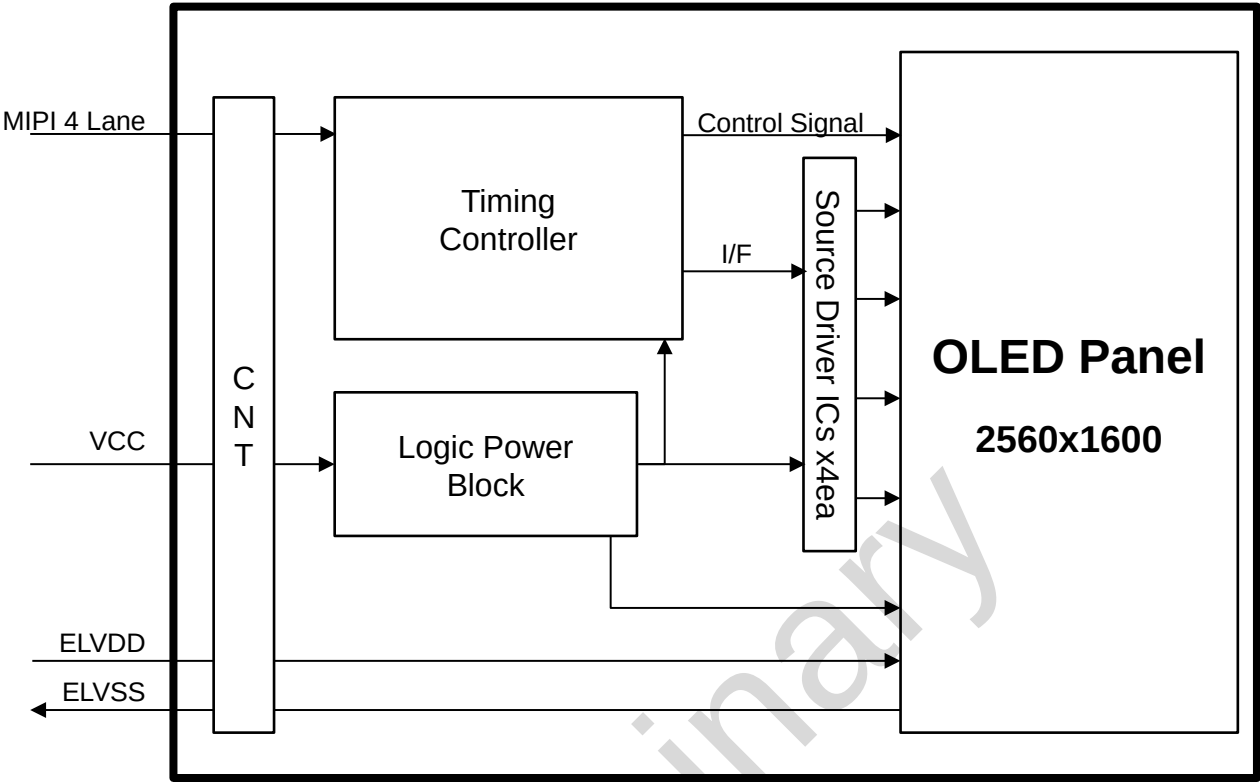
Note (3) Definition of contrast ratio (K)

$$Contrast\ Ratio(K) = \frac{\text{Brightness of selected dot (White patterned area) at } 420cd/m^2}{\text{Brightness of non - selected dot (Black patterned area) at } 420cd/m^2}$$

Note (4) Optical measuring system, temperature regulated chamber
external Light : dark state .
Test angle is +80~-80 degrees due to measuring system limitation.



4. BLOCK DIAGRAM



5. ELECTRICAL CHARACTERISTICS

5.1 OLED MODULE

(T_{amb} = -20 to 60°C)

Item		Symbol	Condition	Min.	Typ.	Max.	Unit	Note
System	Analog/Logic Vol.	VDD_3.3V	-	3.0	3.3	3.6	VDC	
Panel	Analog Vol.	VBAT	-	2.9	-	5.0	VDC	
Input Voltage	H-level	VIH	-	0.7xVDD_1. 8	-	VDD_1.8	VDC	
	L-level	VIL	-	Vss	-	0.3xVDD_1. 8	VDC	
Output Voltage	H-level	VOH	IOH = -2/4/8/12mA	0.8xVDD_1. 8	-	VDD_1.8	VDC	
	L-level	VOL	IOH = -2/4/8/12mA	Vss	-	0.2xVDD_1. 8	VDC	
Current	Logic	IVDD_3.3V	Full White, 60Hz VDD_3.3V=3.3V	-	(324)	-	mA	(1) (2)
	Logic	IVDD_3.3V	Full White, 120Hz VDD_3.3V=3.3V	-	(406)	-		
	Panel	IVBAT	Full White VBAT=3.8V	-	(1380)	-		
Supply Voltage (TSP)	Analog Voltage	TSP_AVDD	-	2.7	3.0	3.4	V	
	Logic Voltage	TSP_DVDD	-		1.8			
Logic Input Voltage (TSP)	“H” Level	VIH	TSP_AVDD=3.0V VDDIO=1.8V or VDDIO=TSP_AVDD	0.75*VDDIO	-	VDDIO+0.3		
	“L” Level	VIL		-0.3	-	0.25*VDDIO		
Logic Output Voltage (TSP)	“H” Level	VOH		0.85*VDDIO	-	-		
	“L” Level	VOL		-	-	0.15*VDDIO		
Frame Frequency		fFRM	-	-	60↔120	-	Hz	(3)

Note(1) : VDD_3.3V=3.3(V), Full White pattern. Temperature = 22±3°C Room Temperature

Note(2) : Measured using SDC Standard Operating Circuit

Current Consumption could be different under customer's DC-DC circuit design.

Note(3) : 60Hz/120Hz Manual Switchable

5.2 Touch Characteristics

- TSP IC : GT6975P
- Firmware : GT6975P_00.00.01.04(7640DFFC).BIN
- Config : AMSA26ZP01(6A04)_160K_20210420(05).cfg

1) Command test

- Attention Test: Pass
- Reset Test: Pass
- T-IC F/W Version Test: Pass
 - ① Main ver.: 0x01
 - ② Sub ver.: 0xEE
- Channel Num. Test: 16, 35
- Lockdown Code Test: Pass

2) Mutual Capacitance Test (Raw) : 4586 ~ 13855 [Each Node AVG ± 27%]

3) Gap Test (Raw Uniformity) : Max 12 (normal), Max 20 (edge), Max 40 (corner)

4) Self Capacitance Test

CB Tx 4500 ~ 12500 / CB Rx 4500 ~ 12500

SCAP Tx 40 ~ 100 / SCAP Rx 15 ~ 60

5) Short Test: Pass

5.3 INTERFACE TIMING

5.3.1 TIMING PARAMETERS

Resolution	2560(H) x 1600(V)		
MIPI Speed	Typ	1500Mbps	
Frame Freq.	Typ (Switchable)	60 Hz	120Hz

5.4 POWER ON/OFF SEQUENCE

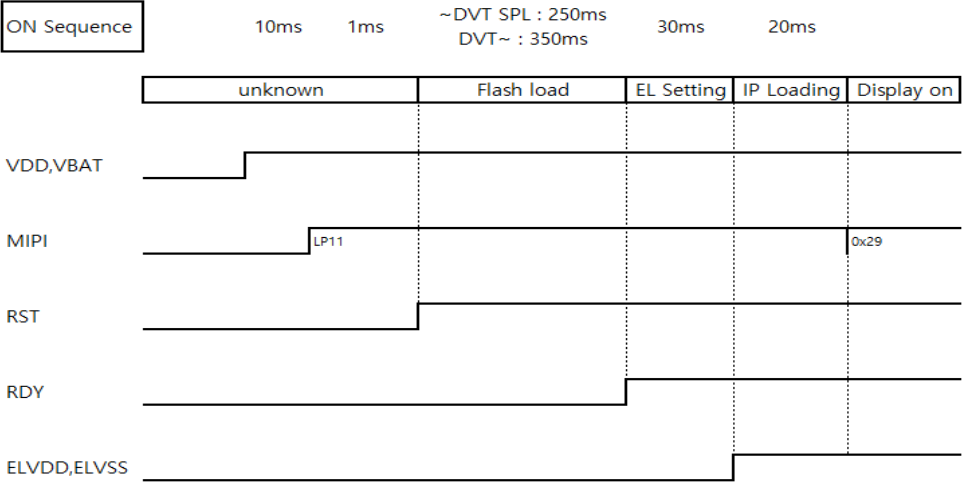
5.4.1 Power On Sequence

	Sequence	Reference
1	Power Off Status	
2	System Power on 1. Turn on VBAT 2. Turn on VDD1	Note1
3	Wait 10ms	Note1
4	MIPI Data, Clk : LP 00 → 01 → 11	Note1
5	Wait 1ms	Note1
6	TCON_RST (L → H)	Note1
7	Wait TCON_RDY (L → H)	Note1
8	Password unlock	Note3
9	DSC Setting	
10	Freq.Setting	
11	Wait 50ms	
12	Checking ESD_ERR_FLG Pin status	Note4
13	Display On(29h)	Note1
14	Brightness Control	
15	Wait TE signal	
16	Data Stream	
17	Display On Status	

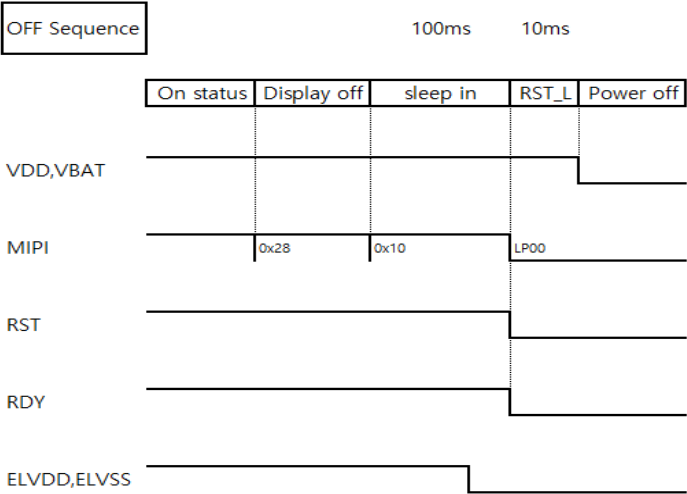
5.4.2 Power Off Sequence

	Sequence	
1	Display On Status	
2	Display Off (28h)	Note2
3	Sleep In (10h)	Note2
4	Wait 100ms	Note2
5	MIPI Data, CLK Lane : LP00 Status	Note2
6	TCON_RST (H → L)	Note2
7	Wait 10ms	Note2
8	System Power Off 1. Turn off VDD1 2. Turn off VBAT	Note2
9	Power Off Status	

Note 1)



Note 2)



Note 3)

Type	Command	R/W	Values	Description
0x39	0xF0	RW	{ 0x5A, 0x5A };	Password unlock
0x39	0xF1	RW	{ 0x5A, 0x5A };	Password unlock
0x39	0xFC	RW	{ 0x5A, 0x5A };	Password unlock

Note 4)

If ESD ERR Flag pin is changed Low, restart power on sequence after power off sequence.
(ESD ERR Flag Low → Power off sequence → Power on sequence)

5.5 INPUT TERMINAL PIN ASSIGNMENT

5.5.1 INPUT SIGNAL (MIPI, Connector : KYOCERA 24-5897-078-002-829+ (PCB Side)

KYOCERA 14-5897-078-002-829+ (Cable Side)

Pin No.	Signal name	Pin No.	Signal name
1	NC	2	NC
3	ELVSS	4	ELVSS
5	ELVSS	6	ELVSS
7	ELVSS	8	ELVSS
9	ELVSS	10	ELVSS
11	NC	12	NC
13	SPI_MISO	14	SPI_CSB
15	SPI_MOSI	16	SPI_CLK
17	INT	18	RST
19	REPORT_EN	20	TSP_AVDD
21	NC	22	TSP_AVDD
23	EL_SCL	24	EL_SDA
25	NC	26	NC
27	VDD_3.3V	28	VDD_3.3V
29	VDD_3.3V	30	VDD_3.3V
31	VDD_3.3V	32	VDD_3.3V
33	NC	34	NC
35	ELVDD	36	ELVDD
37	ELVDD	38	ELVDD
39	ELVDD	40	ELVDD
41	ELVDD	42	ELVDD
43	NC	44	NC
45	GND	46	GND
47	GND	48	D3N
49	GND	50	D3P
51	D2N	52	GND
53	D2P	54	GND
55	GND	56	CLKN
57	GND	58	CLKP
59	D1N	60	GND
61	D1P	62	GND
63	GND	64	D0N
65	GND	66	D0P
67	NC (SDC internal Use)	68	TE
69	NC (SDC internal Use)	70	NC (SDC internal Use)
71	NC (SDC internal Use)	72	NC (SDC internal Use)
73	BIST_EN	74	TCON_RST
75	TCON_RDY	76	TCON_INTR
77	GND	78	GND

6. MODULE PICTURE



< Front Side >



< Rear Side >

Model No.: AMSA26ZP01		SAMSUNG Display CO., LTD. (All rights reserved.)		
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7. RELIABILITY TEST

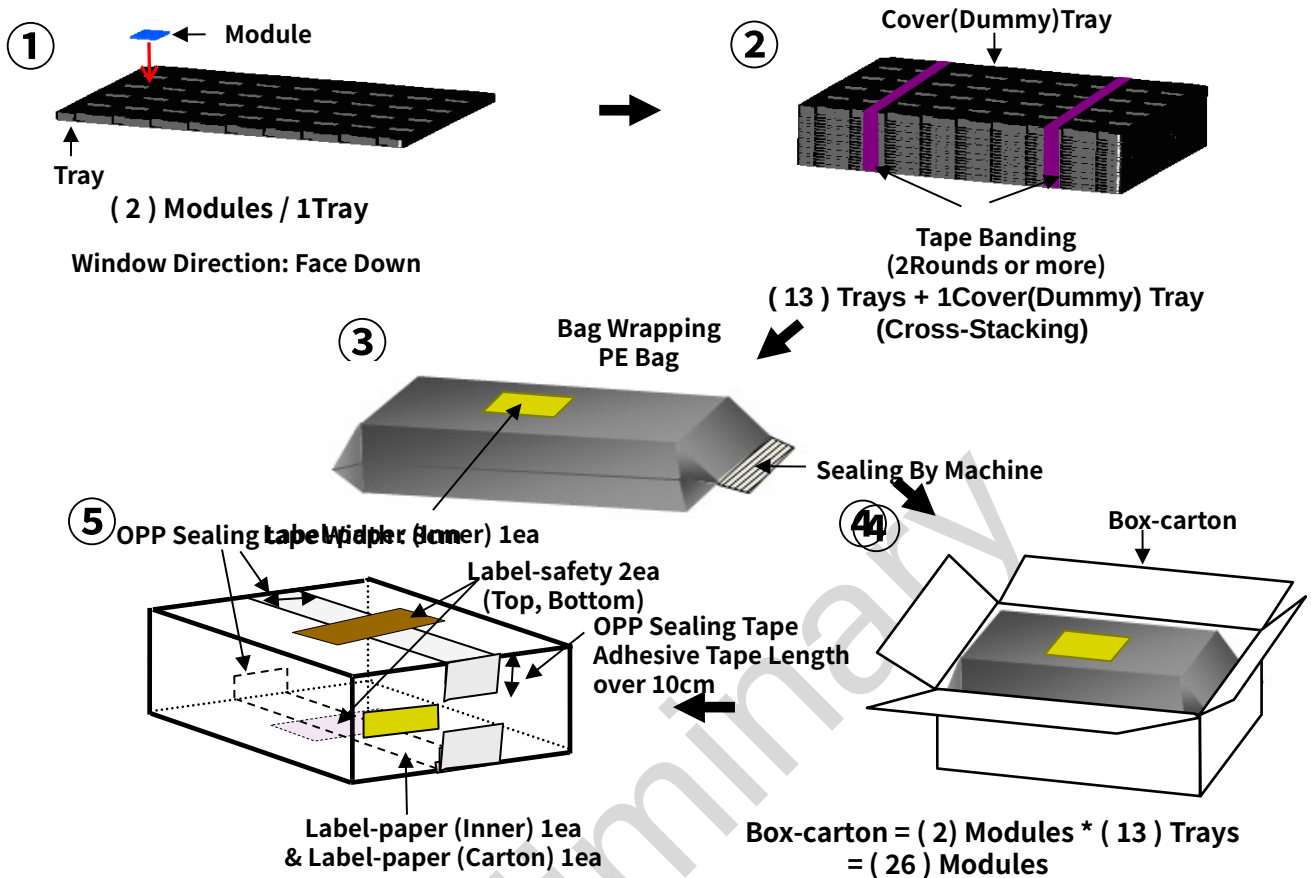
Item		Condition	Time/Cycle
HTOL		70 °C	240 hrs
LTOL		-20 °C	120 hrs
HTS		85 °C	240 hrs
LTS		-40 °C	240 hrs
THB		60 °C / 93%	240 hrs
UHASt		65 °C / 90%	240 hrs
T/C		-40 °C / 85°C	100 cycles
ESD	Non Operating	Contact : Edge ±6kv Center ±8kv	once/point
		Air(non-contact) : Edge ±8kv Center ±15kv,	once/point
Box Vibration (Non-operating)		1.047Grms, 6~200Hz, X/Y/Z 1Hr	1time

[Result Evaluation Criteria]

Under the display quality test conditions with normal operation state, these should be no change which may affect practical display functions.

8. Packing

8.1 Box Pack

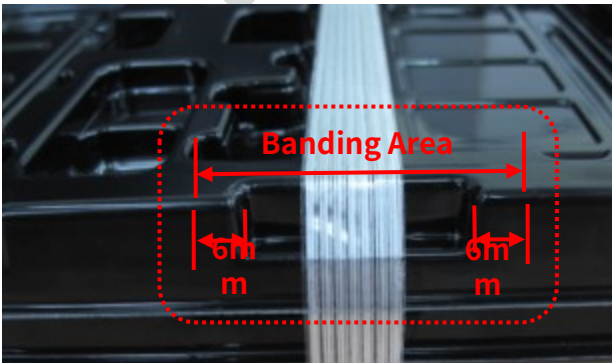
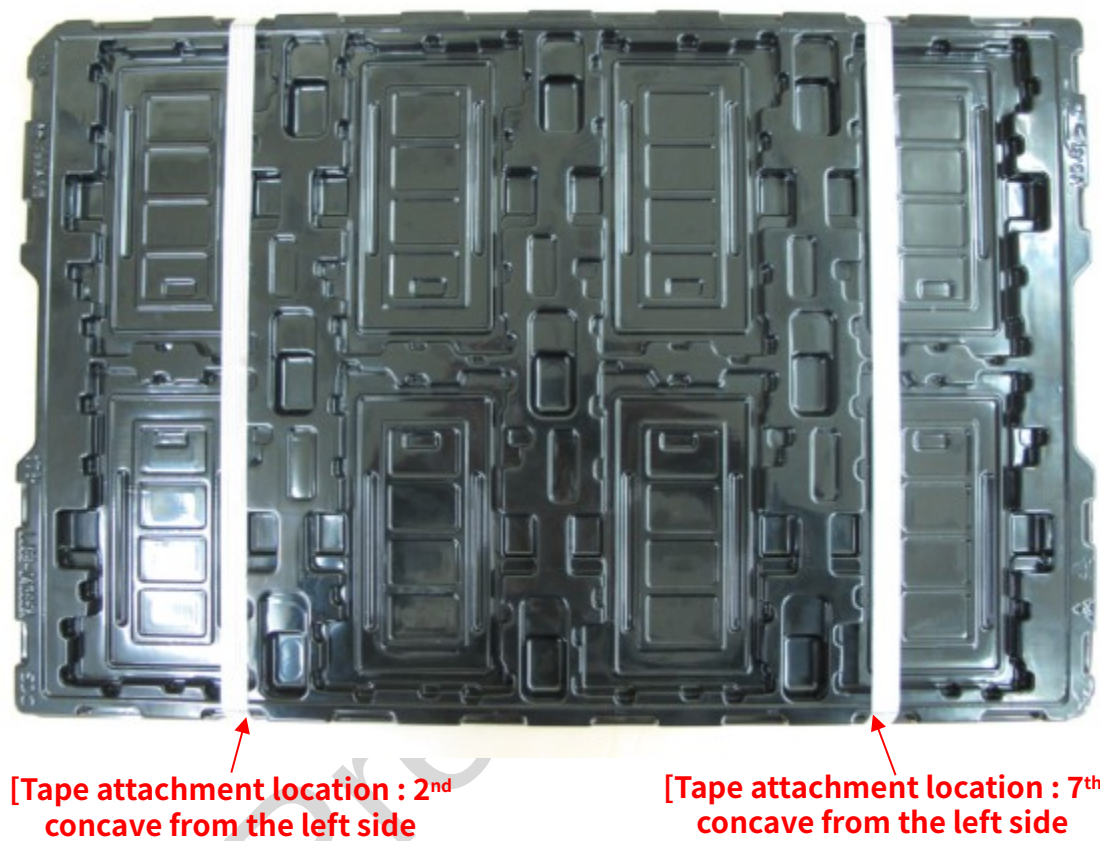


Note

- (1) Total :Box-carton approx. : (10)kg
- (2) Size : 583(L) x 388(W) x 210(H)
- (3) Place the Module in the tray facing the active area direction.
- (4) Stack the trays and cover (dummy) tray.
- (5) Resistance of tray surface : $1 \times 10^5 \sim 1 \times 10^9 \Omega$
 - * Measure Point (Ref. Tray Drawing)
 - Top direction : In pocket, area of placed Module)
 - Bottom direction : In pocket, The nearest "11" shape Stopper with Module)
- (6) Triboelectric Charge of tray surface : Max 100V
 - * Measure Point : In Pocket (Top / Bottom) , 2kgf over 10 times with Clean Wiper
 - * Measurement condition : $22 \pm 3^\circ\text{C} / 50 \pm 5\%$, measure on antistatic mats)
- (7) Wrap the PE bag by packing machine and affix the Label-Paper on Bag.
- (8) Put the bag in the Box-carton .
- (9) Seal the Box-carton and affix the Label-safety & Label-paper.

TRAY Tape Banding (A TYPE TRAY)

- (1) TRAY Tape Banding locations : 2nd & 7th concave from the left side
- (2) TAPE Banding : more than twice

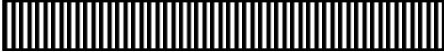
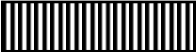
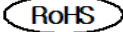


[Banding Area]

8.2 Label

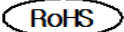
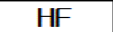
* Label-paper (Inner)

145 mm

		
XXXX-XXXXXXXX-XXXXXXXXXXXXXXXXXX		
CUST CODE : SD18D10893		
Q'TY : XXX PCS /INNER	PO TYPE : E3	
SUPPLYER : SDC /E100-BO		
SDC CODE : AMSA26ZP01 OPT : XX		
LOT NO : XXXXXX		
INSPECTOR : XXX		
DATE : XXXX.XX.XX(XXX) XX,WXX		
note1. note2.		
		
A_	AXXX	ALL SITE
.ETC NAME 1		
.ETC NAME 2		
.ETC NAME 3		
.ETC NAME 4		

102mm

* Label-paper (Carton)

		
XXXX-XXXXXXXX-XXXXXXXXXXXXXXXXXX		
CUST CODE : SD18D10893		
Q'TY : XXX PCS /CARTON	PO TYPE : E3	
SUPPLYER : SDC /E100-BO		
SDC CODE : AMSA26ZP01 OPT : XX		
LOT NO : XXXXXX		
INSPECTOR : XXX		
DATE : XXXX.XX.XX(XXX) XX,WXX		
note1. note2.		
		
A_	AXXX	ALL SITE
.ETC NAME 1		
.ETC NAME 2		
.ETC NAME 3		
.ETC NAME 4		

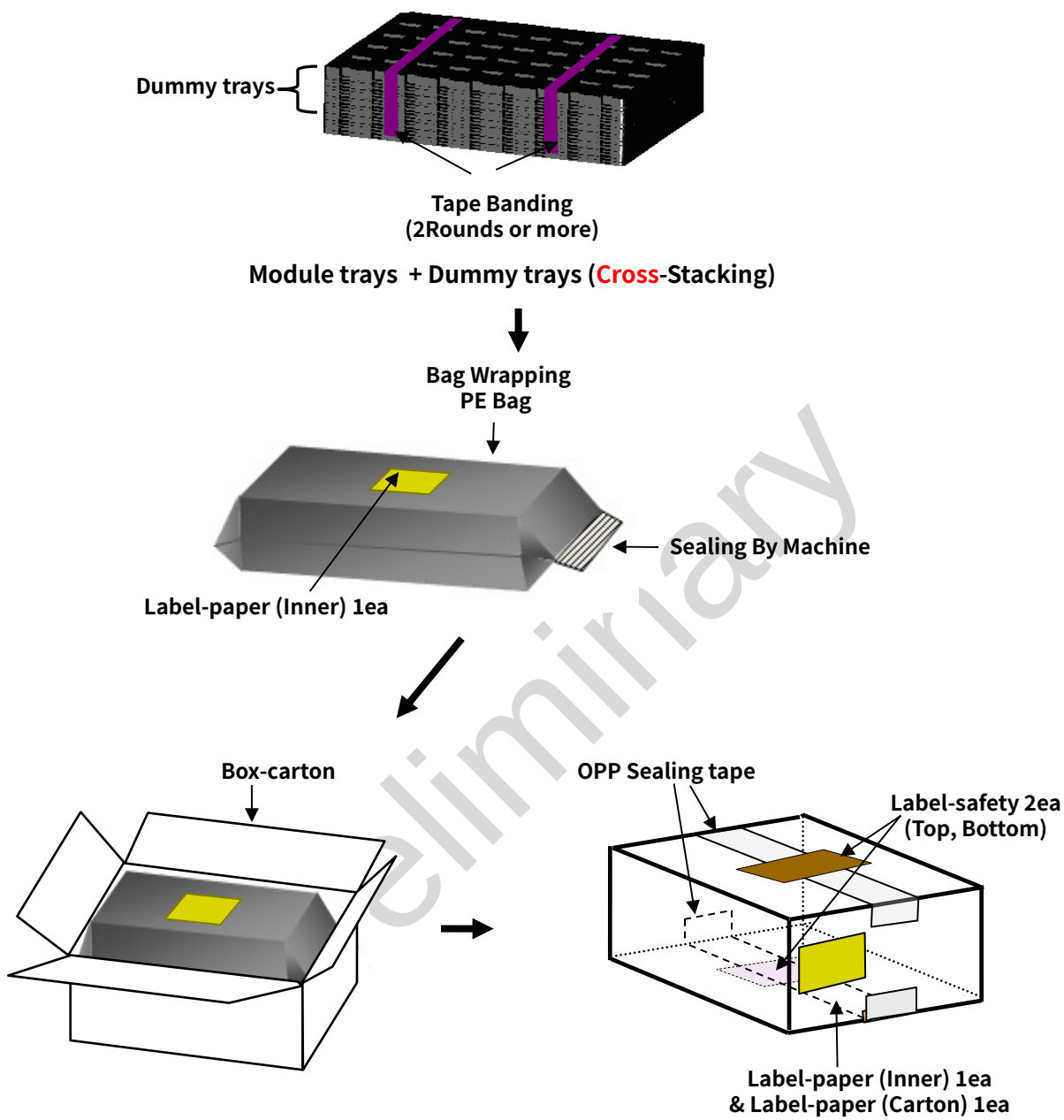
* Label-safety

80mm

30mm

TO BE REMOVED BY USER ONLY If this seal is broken or tampered before use, please call collect the below number.	
TELEPHONE NUMBER 02-1234-5678	MANUFACTURED IN CHINA DATE: 2015.05.15

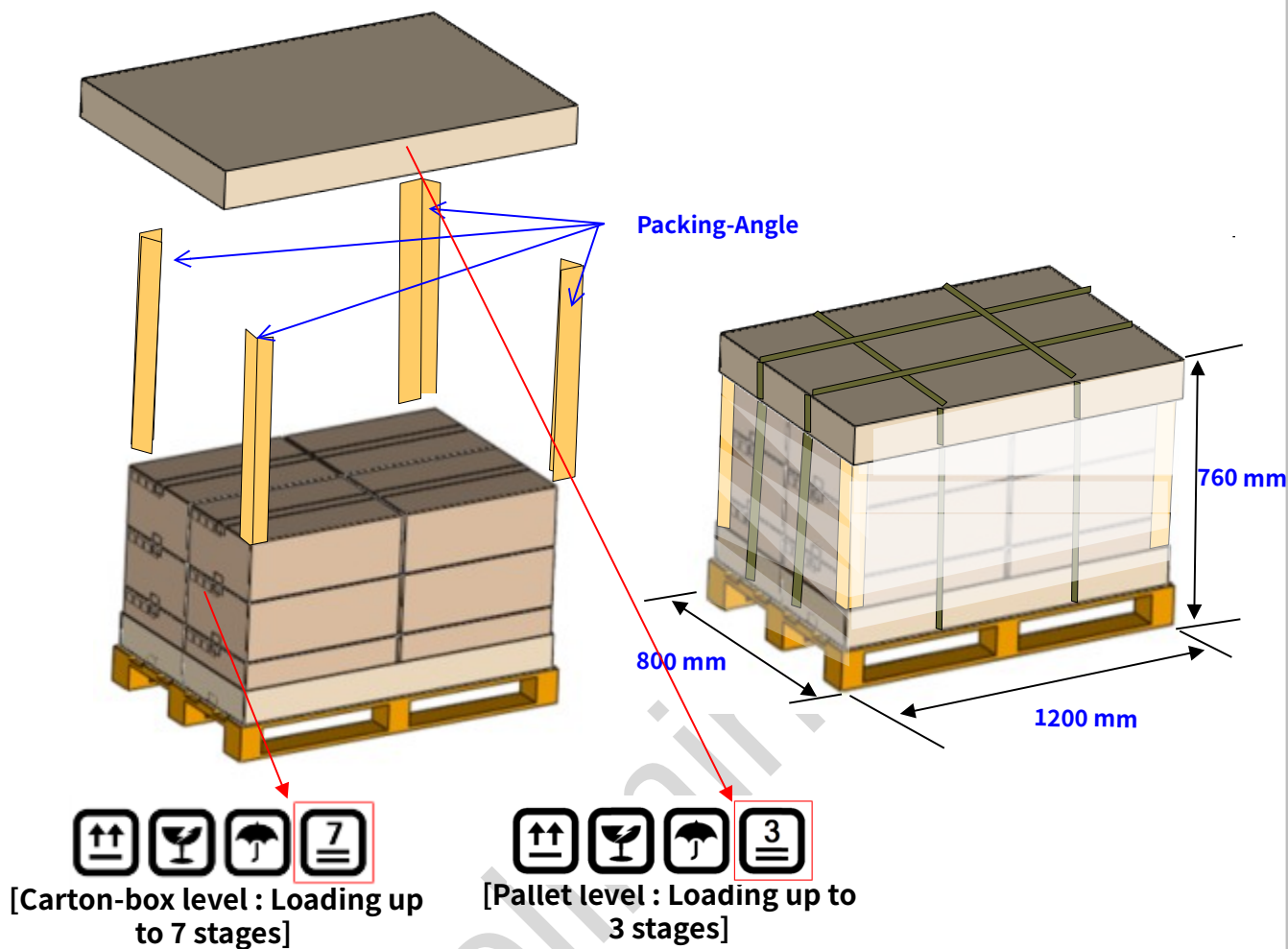
8.3 Packing for Small Quantities



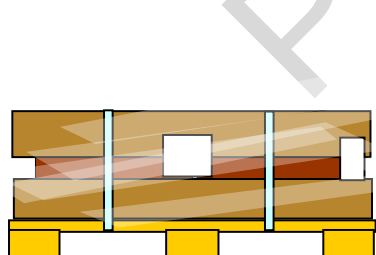
Note

When package quantity is small, Modules containing trays are stacked the bottom, and dummy trays are stacked at the top of package, then wrap the PE bag by Packing machine and affix the Label-Paper on Bag. Put the Bag in the Box-carton Seal the Box-carton and affix the Label-safety & Label-Paper.

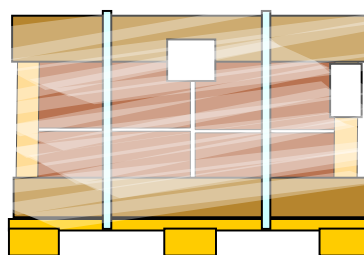
8.4 Over Pack



※ Packing spec. for small quantities



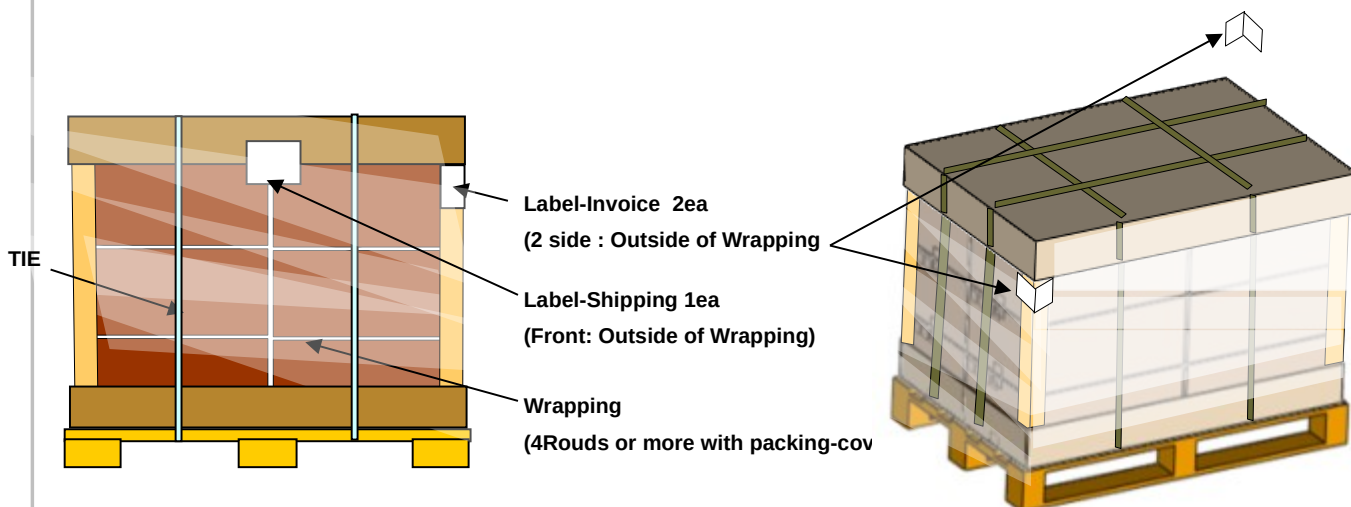
1 step
[Carton Box : 1~4 Box]



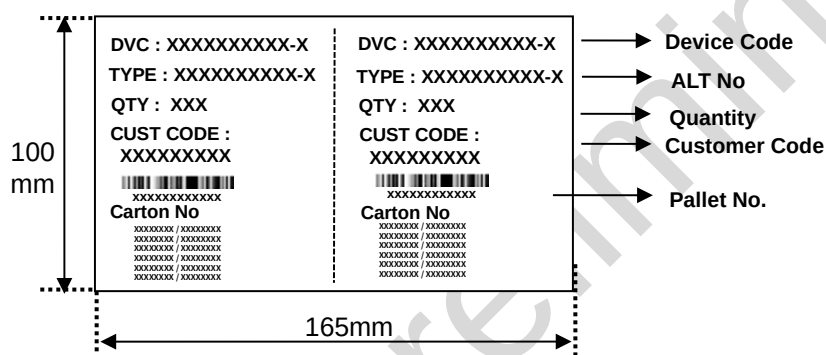
2 step
[Carton Box : 5~8 Box]

- Small Quantities (1step ,2 step) must stack on the Top.

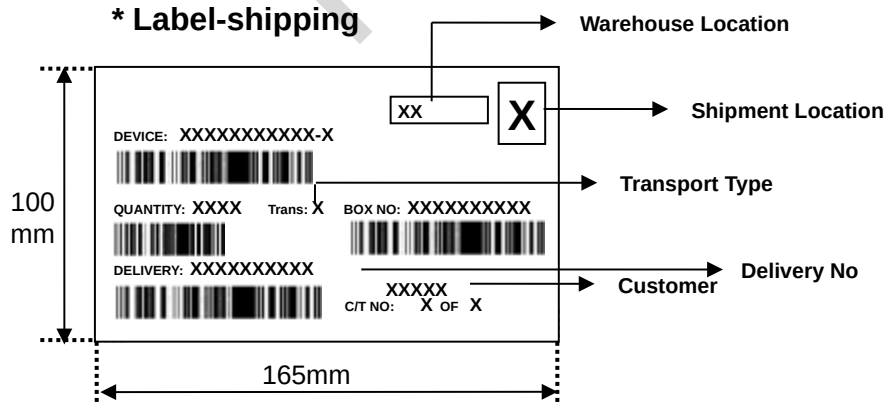
8.5 Over Pack Attach



*** Label-invoice**



*** Label-shipping**



* Reference Image (Except Label and Wrap)

Carton Box		
Pallet		

Caution

For keeping Safe quality from outer exposure or contamination, modules should be consumed with in 2 months after unpacking.

Preliminary

9. GENERAL PRECAUTIONS

9.1 Mounting Method

The AMOLED Panel of SAMSUNG Display CO.,LTD. module consists of two slim glasses with polarizer which can easily get damaged. Since the module is constructed as to be fixed by utilizing fitting holes in the printed circuit board. Extreme care should be used when handling the AMOLED modules.

9.2 Caution of AMOLED Handling and Cleaning

When cleaning the display surface, use soft cloth solvent as recommended below and wipe gently.

- Ⓢ Isopropyl alcohol
- Ⓢ Ethyl alcohol
- Ⓢ Trichlorotrifluoroethane

Do not wipe the display surface with dry or hard materials that will damage the polarizer surface.

Do not use the following solvent.

- Ⓢ Water
- Ⓢ Ketone
- Ⓢ Aromatics

Do not wipe ITO pad area with the dry or hard materials that will damage the ITO patterns.

Do not use the following solvent on the pad and prevent it from being contaminated.

- Ⓢ HCFC
- Ⓢ Soldering flux
- Ⓢ Chlorine(Cl), Sulfur(S)
- Ⓢ Spit, Fingerprint

If the product is not wrapped with a desiccant added pad, ITO pattern can be damaged by corrosion. SAMSUNG Display CO.,LTD. suggests wrapping a product with a desiccant unless customers particularly indicate that they do not want it. In case ITO pattern corrodes due to the usage of chlorine, sulfur or customer's mishandling of the product, the responsibility lies with the customer.

9.3 Caution Against Static Charge

For AMOLED module, use C-MOS LSI drivers, therefore we recommend that you ;

Connect any unused input terminal to VDD or VSS, do not input any signals before power is turned on, and ground your body, work/assembly areas, assembly equipment to protect against static electricity. It could occur static electricity when taping off the film which protects AMOLED.

Against static charge, you should make sure that the product is safe or not by experiment in advance.

9.4 Packing

- Ⓢ The packing principle is that AMOLED module should keep its packing condition at the time of delivery. When storing the AMOLED after unpacking, note the followings.
- Ⓢ AMOLED module is consisted of GLASS and assemblies. It should avoid pressure, strong impact, and being dropped from a height.
- Ⓢ To prevent modules from degradation, do not operate or store them in a place where they are directly exposed to sunlight or high temperature/humidity.

9.5 Caution for Operation

- ◎ If you do not follow normal POWER ON , OFF sequence or abnormal operating, then AMOLED module can be damaged Electro-optically and does not recover.
- ◎ Response time may extremely delay at a temperature lower than operating range, AMOLED does not normally operate at a high temperature. But this may recover at a proper temperature.
- ◎ When you set optimal operating voltage to AMOLED module, you can see the optimal contrast of AMOLED. So, add voltage controllable function at SET Module.
- ◎ AMOLED module may not display normally when twisting power or pressing power is added. Therefore you should secure AMOLED module maximum thickness at set assembly not to have any pressure affect AMOLED module.
- ◎ Electro-chemical reaction may occur when there is humidity on pad, therefore, you should use AMOLED Module below maximum operating humidity.
- ◎ AMOLED Module Power Vdd should be designed to protect surge current at SET Module.
- ◎ You should not damage connector and cable for AMOLED module assembly by force folding or by applying extreme power.
- ◎ AMOLED may not display normally when it is interfered by surrounding elements, therefore you should consider setting design not to damage AMOLED module by surrounding elements.
- ◎ To satisfy EMI standards, you should plan your design after considering emitting energy.
- ◎ We can not guarantee display characteristics outside viewing area, therefore your set window should be fixed into viewing area.
- ◎ Image-sticking may occur if AMOLED displays same image for a long time, so you need to make a pattern change for AMOLED.,

9.6 Storage

- ◎ Place in a dark place where neither exposure to direct sunlight or any fluorescent light is permitted and keep at room temperature & room humidity.
- ◎ Store with no contact with polarizer surface.
It is recommended to store them as they have been contained in the inner container when we delivered them.

9.7 Safety Precautions

- ◎ Disassembly or modification may cause electric shock, damages to sensitive part inside of the AMOLED module, dust adhesion, or scratches on the display part.
- ◎ In the event that the contents of AMOLED module are on skin, wipe them with a paper towel or gauge and wash the part well, and receive medical attention if necessary.
- ◎ Do not use the AMOLED module for the Special purpose besides display units.
- ◎ Be careful of the glass chips that may cause injury to fingers of skin, when the display part is broken.

9.8 Precautions before Use

You should discuss the following case with SAMSUNG Display CO.,LTD.

- ◎ in case of any questions about contents of this "Specification For Approval".
- ◎ in case of occurring new problems not mentioned at this "Specification For Approval".
- ◎ in case of your request about income inspection Specification change.
- ◎ in case of occurring new problem at your driving test.

※ If SDC has to change the conditions Specified in the Specification, previously the negotiation shall be held and decided.

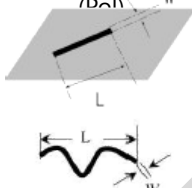
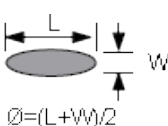
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10. COSMETIC (Inspection Specification)

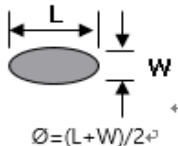
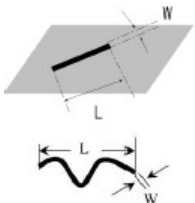
- Inspection Condition

- Viewing distance : 35 ± 5 cm
- Ambient luminance : Function : 200 ± 50 Lux / Appearance : 1000 ± 200 Lux
- Viewing angle: $90^\circ \pm 45^\circ$ (Up/Down/Left/Right), Judged by right angle (90°) to the OLED module surface
- All Items are inspected with Polarizer Protect Film

① Display (Visual Defect & Cosmetic Defect)

No.	Item	Criterion for Defect	Defect Type												
1	No Display	Disallowance	Major												
2	Irregular Display	Disallowance	Major												
3	Line Defect	Disallowance (Vertical line / Horizontal line)	Major												
4	Dot Defect	<table><tr><th colspan="2">Acceptable number</th></tr><tr><th>Dark Dot</th><th>Bright Dot</th></tr><tr><td>9 (DS>5mm)</td><td>0</td></tr><tr><td colspan="2">- Dark dot number should be counted on R,G,B pattern. - 2 adjacent dark dot : $n \leq 1$ (DS>5mm)</td></tr></table>	Acceptable number		Dark Dot	Bright Dot	9 (DS>5mm)	0	- Dark dot number should be counted on R,G,B pattern. - 2 adjacent dark dot : $n \leq 1$ (DS>5mm)		Minor				
Acceptable number															
Dark Dot	Bright Dot														
9 (DS>5mm)	0														
- Dark dot number should be counted on R,G,B pattern. - 2 adjacent dark dot : $n \leq 1$ (DS>5mm)															
5	Scratch/ Line type Defect 	<table><tr><th>Width (mm)</th><th>Length (mm)</th><th>Acceptable number</th></tr><tr><td>$W<0.05$</td><td>Ignore</td><td>Ignore</td></tr><tr><td>$0.05\leq W<0.1$</td><td>$0.3\leq L\leq 3.0$</td><td>4 (DS≥ 30mm)</td></tr><tr><td colspan="2">$0.1\leq W$ or $3.0<L$</td><td>0</td></tr></table> ※ Removable stain or particles with cleaning solution & soft touch	Width (mm)	Length (mm)	Acceptable number	$W<0.05$	Ignore	Ignore	$0.05\leq W<0.1$	$0.3\leq L\leq 3.0$	4 (DS ≥ 30 mm)	$0.1\leq W$ or $3.0<L$		0	Minor
Width (mm)	Length (mm)	Acceptable number													
$W<0.05$	Ignore	Ignore													
$0.05\leq W<0.1$	$0.3\leq L\leq 3.0$	4 (DS ≥ 30 mm)													
$0.1\leq W$ or $3.0<L$		0													
6	Circle type Defect (Pol)  $\varnothing=(L+W)/2$	<table><tr><th>Size Ø (mm)</th><th>Acceptable number</th></tr><tr><td>$\varnothing < 0.20$</td><td>Ignore</td></tr><tr><td>$0.20 \leq \varnothing < 0.5$</td><td>5 (DS ≥ 30mm)</td></tr><tr><td>$0.5\leq \varnothing$</td><td>0</td></tr></table> ※ Removable stain or particles with cleaning solution & soft touch	Size Ø (mm)	Acceptable number	$\varnothing < 0.20$	Ignore	$0.20 \leq \varnothing < 0.5$	5 (DS ≥ 30 mm)	$0.5\leq \varnothing$	0	Minor				
Size Ø (mm)	Acceptable number														
$\varnothing < 0.20$	Ignore														
$0.20 \leq \varnothing < 0.5$	5 (DS ≥ 30 mm)														
$0.5\leq \varnothing$	0														
7	ELA Line Stain	If needed, judged by limit samples. @ 127Gray (420nits)	Minor												
8	Color Difference	If needed, judged by limit samples. @ 127Gray / 255Gray (420nits)	Minor												
9	Random Mura	If needed, judged by limit samples. @ 127Gray (420nits)	Minor												

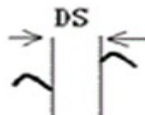
② Window

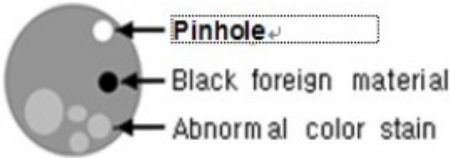
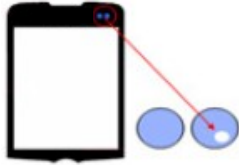
No.	Item	Criterion for Defect	Defect Type																									
1	Window Circular Particle (White/Black) 	- V/A Area(Viewing Area) <table><tr><th>Size Ø (mm)</th><th>Acceptable Number</th></tr><tr><td>Ø ≤ 0.05</td><td>Ignore</td></tr><tr><td>0.05 < Ø ≤ 0.2</td><td>Ignore (D S ≥ 10mm)</td></tr><tr><td>0.2 < Ø ≤ 0.4</td><td>4 (D S ≥ 10mm)</td></tr></table> - P/A(Printing Area) <table><tr><th>Size Ø (mm)</th><th>Acceptable Number</th></tr><tr><td>Ø ≤ 0.2</td><td>Ignore</td></tr><tr><td>0.2 < Ø ≤ 0.4</td><td>6 (D S ≥ 10mm)</td></tr></table> ※ Removable stain or particles with cleaning solution & soft touch are allowed	Size Ø (mm)	Acceptable Number	Ø ≤ 0.05	Ignore	0.05 < Ø ≤ 0.2	Ignore (D S ≥ 10mm)	0.2 < Ø ≤ 0.4	4 (D S ≥ 10mm)	Size Ø (mm)	Acceptable Number	Ø ≤ 0.2	Ignore	0.2 < Ø ≤ 0.4	6 (D S ≥ 10mm)	Minor											
Size Ø (mm)	Acceptable Number																											
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Size Ø (mm)	Acceptable Number																											
Ø ≤ 0.2	Ignore																											
0.2 < Ø ≤ 0.4	6 (D S ≥ 10mm)																											
2	Window Dent/ Bubble	- V/A Area(Viewing Area) <table><tr><th>Size Ø (mm)</th><th>Acceptable Number</th></tr><tr><td>Ø ≤ 0.10</td><td>Ignore</td></tr><tr><td>0.10 < Ø ≤ 0.15</td><td>Ignore (D S ≥ 10mm)</td></tr><tr><td>0.15 < Ø ≤ 0.30</td><td>3 (D S ≥ 10mm)</td></tr></table> - P/A(Printing Area) <table><tr><th>Size Ø (mm)</th><th>Acceptable Number</th></tr><tr><td>Ø ≤ 0.10</td><td>Ignore</td></tr><tr><td>0.10 < Ø ≤ 0.2</td><td>6 (D S ≥ 10mm)</td></tr><tr><td>0.2 < Ø ≤ 0.30</td><td>2 (D S ≥ 10mm)</td></tr></table> <table><tr><th>Width (mm)</th><th>Length (mm)</th><th>Acceptable Number</th></tr><tr><td>W ≤ 0.02</td><td>Ignore</td><td>Ignore</td></tr><tr><td>0.02 < W ≤ 0.08</td><td>L ≤ 3</td><td>4 (D S ≥ 5mm)</td></tr></table>	Size Ø (mm)	Acceptable Number	Ø ≤ 0.10	Ignore	0.10 < Ø ≤ 0.15	Ignore (D S ≥ 10mm)	0.15 < Ø ≤ 0.30	3 (D S ≥ 10mm)	Size Ø (mm)	Acceptable Number	Ø ≤ 0.10	Ignore	0.10 < Ø ≤ 0.2	6 (D S ≥ 10mm)	0.2 < Ø ≤ 0.30	2 (D S ≥ 10mm)	Width (mm)	Length (mm)	Acceptable Number	W ≤ 0.02	Ignore	Ignore	0.02 < W ≤ 0.08	L ≤ 3	4 (D S ≥ 5mm)	Minor
Size Ø (mm)	Acceptable Number																											
Ø ≤ 0.10	Ignore																											
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Width (mm)	Length (mm)	Acceptable Number																										
W ≤ 0.02	Ignore	Ignore																										
0.02 < W ≤ 0.08	L ≤ 3	4 (D S ≥ 5mm)																										
3		- V/A Area(Viewing Area) <table><tr><th>Width (mm)</th><th>Length (mm)</th><th>Acceptable Number</th></tr><tr><td>W ≤ 0.02</td><td>Ignore</td><td>Ignore</td></tr><tr><td>0.02 < W ≤ 0.03</td><td>L ≤ 5</td><td>Ignore (D S ≥ 5mm)</td></tr><tr><td>0.03 < W ≤ 0.08</td><td>L ≤ 5</td><td>4 (D S ≥ 5mm)</td></tr></table> - P/A(Printing Area) <table><tr><th>Width (mm)</th><th>Length (mm)</th><th>Acceptable Number</th></tr><tr><td>W ≤ 0.02</td><td>Ignore</td><td>Ignore</td></tr><tr><td>0.02 < W ≤ 0.03</td><td>L ≤ 5</td><td>Ignore (D S ≥ 5mm)</td></tr><tr><td>0.03 < W ≤ 0.08</td><td>L ≤ 5</td><td>4 (D S ≥ 5mm)</td></tr></table> ※ Measure the thickness based on the thickest part of the linear foreign object. ※ If the linear foreign particle is bundled or twisted, apply the circular foreign object standard. ※ Allow contamination and stains that are removed by lightly wiping with soft cloth	Width (mm)	Length (mm)	Acceptable Number	W ≤ 0.02	Ignore	Ignore	0.02 < W ≤ 0.03	L ≤ 5	Ignore (D S ≥ 5mm)	0.03 < W ≤ 0.08	L ≤ 5	4 (D S ≥ 5mm)	Width (mm)	Length (mm)	Acceptable Number	W ≤ 0.02	Ignore	Ignore	0.02 < W ≤ 0.03	L ≤ 5	Ignore (D S ≥ 5mm)	0.03 < W ≤ 0.08	L ≤ 5	4 (D S ≥ 5mm)	Minor	
Width (mm)	Length (mm)	Acceptable Number																										
W ≤ 0.02	Ignore	Ignore																										
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Width (mm)	Length (mm)	Acceptable Number																										
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No.	Item	Criterion for Defect	Defect Type																			
4	Window Scratch 	- V/A Area(Viewing Area)	Minor																			
		<table><tr><th>Width (mm)</th><th>Length (mm)</th><th>Acceptable Number</th></tr><tr><td>$W \leq 0.02$</td><td>Ignore</td><td>Ignore</td></tr><tr><td>$0.02 < W \leq 0.08$</td><td>$L \leq 5$</td><td>3 (DS ≥ 5mm)</td></tr></table>		Width (mm)	Length (mm)	Acceptable Number	$W \leq 0.02$	Ignore	Ignore	$0.02 < W \leq 0.08$	$L \leq 5$	3 (DS ≥ 5 mm)										
		Width (mm)		Length (mm)	Acceptable Number																	
		$W \leq 0.02$		Ignore	Ignore																	
		$0.02 < W \leq 0.08$		$L \leq 5$	3 (DS ≥ 5 mm)																	
		- P/A(Printing Area)		<table><tr><th>Width (mm)</th><th>Length (mm)</th><th>Acceptable Number</th></tr><tr><td>$W \leq 0.02$</td><td>Ignore</td><td>Ignore</td></tr><tr><td>$0.02 < W \leq 0.03$</td><td>$L \leq 5$</td><td>Ignore (DS ≥ 5mm)</td></tr><tr><td>$0.03 < W \leq 0.08$</td><td>$L \leq 5$</td><td>4 (DS ≥ 5mm)</td></tr></table>	Width (mm)	Length (mm)	Acceptable Number	$W \leq 0.02$	Ignore	Ignore	$0.02 < W \leq 0.03$	$L \leq 5$	Ignore (DS ≥ 5 mm)	$0.03 < W \leq 0.08$	$L \leq 5$	4 (DS ≥ 5 mm)						
		Width (mm)			Length (mm)	Acceptable Number																
		$W \leq 0.02$			Ignore	Ignore																
		$0.02 < W \leq 0.03$			$L \leq 5$	Ignore (DS ≥ 5 mm)																
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Width (mm)	Length (mm)	Acceptable Number																				
$W \leq 0.02$	Ignore	Ignore																				
$0.02 < W \leq 0.03$	$L \leq 5$	Ignore (DS ≥ 5 mm)																				
$0.03 < W \leq 0.08$	$L \leq 5$	4 (DS ≥ 5 mm)																				
5	Window Chip-ping	<table><tr><th>Size (mm)</th><th>Length (mm)</th><th>Acceptable Number</th></tr><tr><td>$W < 0.1$</td><td>$L \leq 0.5$</td><td>Ignore</td></tr><tr><td>$0.1 < W \leq 0.2$</td><td>$L \leq 0.5$</td><td>2</td></tr><tr><td>$0.2 < W$</td><td>$0.5 < L$</td><td>0</td></tr></table>	Size (mm)	Length (mm)	Acceptable Number	$W < 0.1$	$L \leq 0.5$	Ignore	$0.1 < W \leq 0.2$	$L \leq 0.5$	2	$0.2 < W$	$0.5 < L$	0	Minor							
		Size (mm)	Length (mm)	Acceptable Number																		
		$W < 0.1$	$L \leq 0.5$	Ignore																		
		$0.1 < W \leq 0.2$	$L \leq 0.5$	2																		
		$0.2 < W$	$0.5 < L$	0																		
6	Edge light leak-age	<table><tr><th>Size (mm)</th><th>Length (mm)</th><th>Acceptable Number</th></tr><tr><td>$W < 0.1$</td><td>$L \leq 20$</td><td>Ignore</td></tr><tr><td>$0.1 < W$</td><td>$20 < L$</td><td>0</td></tr></table>	Size (mm)	Length (mm)	Acceptable Number	$W < 0.1$	$L \leq 20$	Ignore	$0.1 < W$	$20 < L$	0	Minor										
		Size (mm)	Length (mm)	Acceptable Number																		
		$W < 0.1$	$L \leq 20$	Ignore																		
		$0.1 < W$	$20 < L$	0																		
7	Window Ink / Printing	- Printing Inside / Outside	Minor																			
		<table><tr><th>Area</th><th>Size Ø (mm)</th><th>Length (mm)</th><th>Acceptable Number</th></tr><tr><td>Pin Hole/ Splashes</td><td colspan="3">Same spec as no#1 window particle</td></tr><tr><td rowspan="2">Saw tooth</td><td>$\text{Ø} \leq 0.10$</td><td>-</td><td>Ignore</td></tr><tr><td>$0.10 < \text{Ø} \leq 0.20$</td><td>DS$\geq 10$</td><td>2</td></tr><tr><td>Wrong, Fuzzy, Incomplete Printing</td><td colspan="3">Not allowable</td></tr></table>		Area	Size Ø (mm)	Length (mm)	Acceptable Number	Pin Hole/ Splashes	Same spec as no#1 window particle			Saw tooth	$\text{Ø} \leq 0.10$	-	Ignore	$0.10 < \text{Ø} \leq 0.20$	DS ≥ 10	2	Wrong, Fuzzy, Incomplete Printing	Not allowable		
		Area		Size Ø (mm)	Length (mm)	Acceptable Number																
		Pin Hole/ Splashes		Same spec as no#1 window particle																		
		Saw tooth		$\text{Ø} \leq 0.10$	-	Ignore																
				$0.10 < \text{Ø} \leq 0.20$	DS ≥ 10	2																
		Wrong, Fuzzy, Incomplete Printing		Not allowable																		

No.	Item	Criterion for Defect	Defect Type															
8	Window (IR LED,Smart Vision, TOF, ALS Sensor)	<table><tr><td></td><td>Size D (mm)</td><td>Acceptable Number</td></tr><tr><td>Particle(Black) / IR Stain</td><td>$D \leq 0.2$</td><td>1</td></tr><tr><td>Saw tooth edge</td><td>$0.10 < D \leq 0.20$</td><td>2 (DS $\geq 10\text{mm}$)</td></tr></table>  ※ Inspection Condition : Black Back Ground / Reflective Mode ※ Removable stain or particles with cleaning solution & soft touch are allowed ※ Reflection inspection after covering black area under window (picture) ※ ... or Black Ink Print glass or Black ... background : OK)  <table><tr><th>Size (mm)</th><th>Acceptable Number</th></tr><tr><td>$\varnothing \leq 0.1$</td><td>Ignore</td></tr><tr><td>$0.1 < \varnothing \leq 0.15$</td><td>1</td></tr></table>		Size D (mm)	Acceptable Number	Particle(Black) / IR Stain	$D \leq 0.2$	1	Saw tooth edge	$0.10 < D \leq 0.20$	2 (DS $\geq 10\text{mm}$)	Size (mm)	Acceptable Number	$\varnothing \leq 0.1$	Ignore	$0.1 < \varnothing \leq 0.15$	1	Minor
			Size D (mm)	Acceptable Number														
		Particle(Black) / IR Stain	$D \leq 0.2$	1														
		Saw tooth edge	$0.10 < D \leq 0.20$	2 (DS $\geq 10\text{mm}$)														
Size (mm)	Acceptable Number																	
$\varnothing \leq 0.1$	Ignore																	
$0.1 < \varnothing \leq 0.15$	1																	
9	Window Camera Hole	White/Black/Color dot	Minor															
		- Scratch, Stain, Dent, ink deleted are not allowable - saw edge : applying the same spec as printing (inside) ※ Removable stain or particles with cleaning solution & soft touch are allowed ※ Mode : Reflective / Transmissive ※ Background : White / Black																
10	Surface Contamination	- Removable stain or particles with cleaning solution & soft touch are allowed	Minor															
11	Cover PNL Bubbles/ Dent/Stain Particle/ Wrinkle/Scratch	- If the defect is not visible at Front side, these are allowed	Minor															
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12	Cover PNL Tilt	- C-panel should go out of the panel edge	Minor															

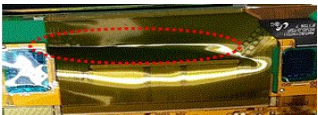
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③ Others

No.	Item	Criterion for Defect	Defect Type
1	Resin Overflow	Overflow & Burr is allowable within 0.3mm	Minor
2	PCB / FPC / COF lifting	- Allow 2.0mm or less of lifting and bubble of M-PCB / T-FPC from C-panel/AL tape - Bridge FPCB : Ignore the lifting	Minor
3	COVER TAPE	1) Wrinkles, scratches, Dent, bubbles, Stab : allowed. However, the following folds are not allowed 2) Tearing : Not allowed 3) No COVER Tape : Not allowed 4) Cover PCB Tape - COVER PCB TAPE covers the PCB and contacts the Cover Panel * Lifting is allowed (If PCB is not visible @ $\Phi=0^\circ$, it is OK) - Ignore C-PCB overlapping with Hook - C-PCB lifting is allowed at COF Bending area (*Do not exceed the bending max)	Minor
4	T-COM Insulation tape	- No insulation tape or Falling off : not allowed - Tape covers the components - Tape is not out of FPCB edge - Bubbles, Wrinkles, Lifting : Allowed	Minor
5	Window Protection Film	1) Window protection film tilt is acceptable 2) Scratches, contamination, wrinkles on the film and bubbles, foreign material between the film and cover glass (* Hole on the PF is not allowed)	Minor
6	COF Bubble	- 8*4mm size is allowed * No quantity limit	Minor
7	COF Folding / Crumple / Wrinkle / Dent	- COF Folding : U Type : OK / Wrinkle / Dent : U Type : allowed, V Type : NG isn  	Minor
8	COF Stain, Contamination	- COF Stain and Contamination are allowed	Minor
9	COF Particle	- Outside of COF Bending (outer side): Removable particle with soft cloth is allowed - COF Bending inside side (inner side): Allow foreign particle less than M-FOB pressure pad 1mm, and other areas are not SIZE	Minor
10	FPC	- Wrinkle, Contamination, Stab, Particle, Bubble, Scratch are allowed if there is no functional problem	Minor

Model No.: AMSA26ZP01

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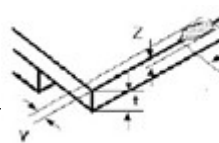
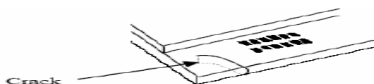
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	Item	Criterion for Defect	Defect Type						
11	PAD Chipping (Pattern Area)	 <table><tr><td>z</td><td>X</td><td>Y</td></tr><tr><td>≤t</td><td>≤5.0mm m</td><td>≤0.4mm</td></tr></table>	z	X	Y	≤t	≤5.0mm m	≤0.4mm	Minor
z	X	Y							
≤t	≤5.0mm m	≤0.4mm							
12	PAD Chipping (No Pattern Area)	 <table><tr><td>z</td><td>X</td><td>Y</td></tr><tr><td>≤t</td><td>≤5.0mm</td><td>≤0.6mm</td></tr></table> 	z	X	Y	≤t	≤5.0mm	≤0.6mm	Minor
z	X	Y							
≤t	≤5.0mm	≤0.6mm							
13	PAD Chipping @Rear Side	<table><tr><td>z</td><td>X</td><td>Y</td></tr><tr><td>≤t</td><td>≤5.0mm</td><td>≤0.5mm</td></tr></table> 	z	X	Y	≤t	≤5.0mm	≤0.5mm	Minor
z	X	Y							
≤t	≤5.0mm	≤0.5mm							
14	Panel Chipping (Side Area)	<table><tr><td>z</td><td>X</td><td>Y</td></tr><tr><td>≤t</td><td>≤3.0mm</td><td>≤0.4mm</td></tr></table> 	z	X	Y	≤t	≤3.0mm	≤0.4mm	Minor
z	X	Y							
≤t	≤3.0mm	≤0.4mm							
15	Panel Crack	- Not allowed	Minor						
16	T-CNT / Bridge / Connector	- CNT should not be broken, apart, crushed, deformed, and etc. - CNT should not be concluded or mis-aligned	Minor						
17	CNT	- CNT damage, Pin dent, Extrusion, Mold damage, Dent , Particles are not allowed	Minor						

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No.	Item	Criterion for Defect	Defect Type
18	QR Code Label	■ QR Code Label Inspection - No Label attachment or Wrong Label : Not allowed - Barcode or text is not out of Label outline - Any removed Barcode or Text is not allowed - Label Tearing or Stab  ed)	Minor
		■ QR Code Label Attachment (X,Y Center) - Cover-Panel should be attached to the open area of the liner - Do not exc - Do not exc - Do not exc 	